

ORIGINAL RESEARCH



Endothelial Elavl1 Is Required for CD8 T-Cell Persistence in Atherosclerosis

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BACKGROUND: Atherosclerotic plaques form through lipid and immune cell accumulation beneath the arterial endothelium. CD8 T cells are abundant in lesions, and their activation correlates with cardiovascular disease severity. Although endothelial cells recruit CD8 T cells to low and disturbed flow regions, whether they regulate CD8 activation and persistence is unknown. These interactions may be mediated by posttranscriptional control of mRNA translation.

METHODS: We used ribosomal profiling in *Cdh5*(PAC); *RiboTRAP* mice to assess posttranscriptional regulation in atherosclerotic endothelium in a partial carotid ligation model. We then used *CreERT2*; *Elavl1* ff; *RiboTRAP* (endothelial cell–knockout) mice to examine the impact of endothelial Elavl1 on gene expression programs and immune cell activity.

RESULTS: Elavl1 motifs were enriched near alternative splicing events, and within 5′ untranslated regions of transcripts with altered ribosomal association in atherosclerosis. Elavl1 suppressed gene expression programs occurring in atherogenic endothelium, as deletion enhanced these responses and reduced CD8 T-cell accumulation (~70%) at plaques without affecting recruitment, consistent with impaired persistence. In vitro, Elavl1-deficient endothelial cells suppressed antigen-dependent CD8 T-cell persistence, even in the presence of wild-type myeloid antigen-presenting cells and cognate antigen.

CONCLUSIONS: Endothelial Elavl1 suppression enables endothelial cells to function as local immune checkpoint–like regulators of CD8 T-cell persistence in atherosclerosis. Thus, endothelial Elavl1 is a key regulator of adaptive immunity.

Key Words: atherosclerosis ■ cardiovascular diseases ■ endothelial cells ■ inflammation ■ mice, knockout

Atherosclerotic lesions are initiated by endothelial cell (EC) activation and dysfunction in areas of arteries exposed to disturbed blood flow, as opposed to steady laminar flow.^{1–3} In response to disturbed flow, changes in aortic EC transcriptional programming facilitate immune cell recruitment and infiltration of the arterial intima.^{4,5} The signals regulating this recruitment process have been well described, and transcriptional networks have been defined in vitro⁶ and more recently in vivo.^{7–9} Major contributors to acute recruitment include adhesion molecules upregulated in ECs (eg, P-selectin, ICAM1 [intercellular adhesion molecule 1], and VCAM [vascular cell adhesion molecular]) and secreted cytokines

(eg, CCL2, CCL5) organized by transcriptional networks including elevated p65/NF- κ B (nuclear factor kappa B) and reduced KLF2,^{10,11} and also cytoskeletal and extracellular matrix changes mediated by integrin α 5 and response-to-injury matrix proteins such as fibronectin.^{12–14} Recruited cells include large numbers of myeloid and lymphoid cells, potentially setting the stage for autoimmune-like damage to the vascular wall. The endothelium is also well positioned to regulate chronic interactions between immune cells and the vessel wall, but very little is known about this process. How ECs respond in these conditions to suppress or enhance immune cell responses is uncharted territory.

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Nonstandard Abbreviations and Acronyms

APC	antigen-presenting cell
BMDC	bone marrow–derived dendritic cell
CLS-1	collagenase type I
EC	endothelial cell
ECKO	postnatal endothelial-specific Elavl1/HuR deletion
FACS	fluorescence-activated cell sorter
FBS	fetal bovine serum
GFP	green fluorescent protein
GM-CSF	granulocyte-macrophage colony-stimulating factor
ICAM1	intercellular adhesion molecule 1
MHC-I	major histocompatibility complex type I
NF-κB	nuclear factor kappa B
OT-I	MHC-I class-restricted, ovalbumen-specific transgenic mice
OVA	ovalbumin
PCAL	partial carotid artery ligation
PCSK9	proprotein convertase subtilisin/kexin type 9
RBP	RNA-binding protein
RT	room temperature
SIINFEKL	processed peptide, antigen for OT-I cells
TCR	T-cell receptor
TRAP	translating ribosome affinity purification
UTR	untranslated region
VCAM	vascular cell adhesion molecule
VEGF	vascular endothelial growth factor
WT	wild-type

What Are the Clinical Implications?

Endothelial activation and the upregulation of adhesion molecules (eg, ICAM1 [intercellular adhesion molecule 1], VCAM-1 [vascular cell adhesion molecule 1]) are essential drivers of acute inflammation and early leukocyte recruitment. The role of the endothelium in regulating chronic T-cell responses within the atherosclerotic plaque remains poorly understood. Intraplaque CD8 T-cell accumulation and activation are strongly associated with increased cardiovascular morbidity and mortality. We identify endothelial cells as active regulators of chronic immune persistence, capable of adopting an immune-suppressive phenotype that limits CD8 T-cell accumulation in inflamed tissues. Using translating ribosome affinity purification, we demonstrate that the RNA-binding protein Elavl1 (HuR) governs the endothelial translome, specifically suppressing the translation of transcripts like C1q and CD27 that modulate T-cell survival. Deletion of endothelial Elavl1 reduces plaque CD8 T-cell levels and activation by >70% across multiple models without affecting systemic circulating levels. We establish a novel long-term coculture system showing that endothelial cells, rather than exogenous cytokines, are sufficient to sustain chronic antigen-specific T-cell expansion. These findings reveal that targeting the endothelial Elavl1-translatome axis offers a precise therapeutic strategy to prune pro-atherogenic T-cell niches, providing a foundation for treating chronic fibro-inflammatory diseases, including atherosclerosis, autoimmunity, and cancer.

they could contribute to EC processes affecting local immune cell activation. However, in addition to regulating mRNA splicing, RBP, and specifically Elavl1/HuR and PTBP1, have been shown to dynamically regulate protein translation in response to cellular stress.^{23,24}

Here, we use translating ribosome affinity purification (TRAP) to examine the chronic adaptation of ECs in the atherosclerotic intima. We identify changes in mRNA accumulation and mRNA splicing at the ribosome. Motifs for Elavl1 are enriched in these mRNA transcripts in the 5' untranslated region (UTR) region important for translational regulation, and the regions flanking alternative spliced domains. Mechanistically, deletion of Elavl1 changed both ribosomal association and mRNA splicing under these conditions, regulating transcriptional programs associated with mitochondrial respiration, translation, and T-cell interactions and immune response. Functionally, loss of Elavl1 in the endothelium reduced local accumulation of CD8 T cells in plaque and reduced responses to antigen in vitro. Together, our data identify a novel stromal checkpoint in atherosclerotic lesions, defined by Elavl1 expression and altered ribosomal association of mRNA.

Understanding whether ECs restrain unwanted immune cell activation is essential for understanding plaque progression, because immune cell infiltration is a common feature of the largest risk factors for cardiovascular disease, including hypercholesterolemia, aging, autoimmune disease, and viral infection. Each of these conditions drives systemic activation of circulating T cells,^{15,16} which are preferentially recruited to sites of disturbed flow.^{17,18} CD8 T effector cells have a prominent role in the plaque, where they are retained after recruitment for months to years, poised for the secretion of cytotoxic and destructive proteins like granzyme K and perforin.^{17,19} Previously, we demonstrated changes in splicing of pre-mRNA encoding proteins involved in mesenchymal-like and immune transitions of ECs under disturbed flow, and identified a large set of regulatory RBP (RNA-binding proteins), including Elavl1/HuR and PTBP1.^{14,20–22} These programs respond directly to the recruitment of myeloid and lymphoid cells to the arterial wall, suggesting that

MATERIALS AND METHODS

Data Availability Statement

All data and materials have been made publicly available at the National Institutes of Health Sequence Read Archive and can be accessed at PRJNA1142064. Data analysis to examine pathway enrichment was performed in Jupyter Notebook, with code outlined here (https://github.com/pamurphyUCONN/2024_Nicholas_Elav1).

Mice

For Elavl1 EC knockout experiments, *Cdh5(PAC)-CreERT2* and *Elavl1f/f* mice were used and have been previously described.^{25,26} They were intercrossed to generate Elavl1 postnatal endothelial-specific Elavl1/HuR deletion (ECKO) mice (*Cdh5(PAC)-CreERT2; Elavl1f/f*) and littermate controls (*Elavl1f/f*) used here. For partial carotid artery ligation (PCAL) and 3-month chronic hyperlipidemia models, Elavl1 excision was accomplished by intraperitoneal injections of either 1 mg (100 μ L of 10 mg/mL) tamoxifen (T5648; Sigma) in sunflower oil 3 $\times$, every other day, at least 1 week before experimental perturbations. Mice were used for experiments in sex-matched and litter-matched pairs.

For TRAP RNAseq experiments, the *Rosa26-fsTRAP* has been previously described.²⁷ *Rosa26-fsTRAP* mice were intercrossed with *Cdh5(PAC)-CreERT2; Elavl1f/f* mice to generate Elavl1 ECKO RiboTRAP mice (*Cdh5(PAC)-CreERT2; Elavl1f/f; Rosa26-fsTRAP*). To generate Elavl1 wild-type (WT) RiboTRAP control mice (*Cdh5(PAC)-CreERT2; Rosa26-fsTRAP*), *Rosa26-fsTRAP* mice were intercrossed with *Cdh5(PAC)-CreERT2*. Tamoxifen was delivered as above.

For the generation of immortalized Elavl1 ECKO mouse aortic ECs, previously described *R26-CreERT2* (*Gt(ROSA)26Sor^{tm1(Cre/ERT2)Tyj/J}*) mice²⁸ were intercrossed with *Elavl1f/f* mice to generate *RosaCreER; Elavl1f/f* experimental mice.

For OT-I (MHC-I class-restricted, ovalbumin-specific transgenic mice) T-cell isolation, OVA (ovalbumin, antigen for OT-I T cells, requiring processing for presentation; SIINFEKL257-264)-specific OT-I TCR (T-cell receptor) transgenic recombination activation gene-1 deficient (*Rag1^{-/-}*) on C57BL/6J mice background were used. Mice were homozygous for CD45.1 or, as previously described,²⁹ WT (CD45.2) bone marrow cells were isolated from C57BL/6J mice.

All mice were housed and handled in accordance with protocols approved by the University of Connecticut Health Center for Comparative Medicine.

Models of Atherosclerosis

For both chronic and PCAL models, hyperlipidemia was induced via 100 μ L intraperitoneal injection containing 1E11 viral particles of AAV8-mPCSK9 (pAAV/D377Y-mPCSK9)^{30,31} produced at the Gene Transfer Vector Core (Schepens Eye Research Institute and Massachusetts Eye and Ear Infirmary, Harvard Medical School). Mice were also placed on the Clinton/Cybulsky high-fat rodent diet with regular casein and 1.25% added cholesterol (D12108C; Research Diets). Hyperlipidemia was confirmed after blood collection via cardiac puncture through the right ventricle. Serum was collected after centrifugation and stored at -80°C until analysis by Total Cholesterol Assay Kit (STA-384; Cell Bio Labs INC).

PCAL was performed as previously described.²⁰ Briefly, although under isoflurane anesthesia, the left external carotid,

internal carotid, and occipital arteries were ligated. Mice were treated postsurgery with Buprenex or Ethiqua as analgesics. High-resolution Doppler ultrasound (Vevo 3100, MX550D 32 MHz transducer) confirmed flow reduction in the partially ligated left carotid artery and not in the contralateral artery. Mice were maintained at a heart rate of ≈ 400 to 500 BPM, and images of Doppler readings were obtained to confirm reduced velocity.

Total Carotid Intimal EC RNA Collection

Elavl1 ECKO mice (*Cdh5(PAC)-CreERT2; Elavl1f/f*) and littermate controls (*Elavl1f/f*) were treated with tamoxifen (3 $\times$ 1 mg, IP) and PCAL and hyperlipidemia induced as described. Total carotid intimal RNA was isolated 3 weeks after PCAL induction. Carotid arteries were dissected and rapidly flushed with 150 μ L of Trizol (15596026; Thermo). Trizol eluant was stored at -80°C until the time of RNA isolation. For RNA isolation, chloroform was added at 1:5 ratio to Trizol (30 μ L:150 μ L; Trizol), and the clear supernatant containing mRNA was combined 1:1 with 70% ethanol (EtOH) and then run on Qiagen RNeasy Micro Kit (74004; Qiagen).

TRAP RNA Collection



Elavl1 ECKO RiboTRAP mice (*Cdh5(PAC)-CreERT2; Elavl1f/f; Rosa26-fsTRAP*) and controls (*Cdh5(PAC)-CreERT2; Rosa26-fsTRAP*) were treated with tamoxifen for recombination and then subjected to PCAL and hyperlipidemia, as described above. Carotid TRAP RNA was isolated 3 weeks after PCAL induction. Mice were perfused through the left ventricle with cold PBS. Carotid arteries were dissected and flushed with cold Lysis Buffer (20 mM RNase-free HEPES potassium hydroxide pH7.4, 5 mM RNase-free Magnesium Chloride, 150 mM RNase-free potassium chloride, 0.5 mM dithiothreitol, complete protease inhibitor, EDTA-free, 100 μ g/mL cycloheximide in methanol, 40 U/mL RNase inhibitor, 1% NP-40, 30 mmol/L 1,2-diheptanoyl-sn-glycero-3-phosphocholine in RNase-free PBS) to extract intimal protein and RNA.

GFP (green fluorescent protein)–(HtzGFP_04, clones 19C8 and clone19F7, Memorial Sloan Kettering Antibody and Biosource Core) or IgG– (15006; Sigma) conjugated M270 Dynabeads (14301; Fisher) were prepared per manufacturer's instructions. Lysate was then incubated with 1 mg of conjugated beads per vessel at 4°C for 30 minutes. After immunoprecipitation for 1 minute at 4°C , the supernatant was collected and combined with RNA lysis buffer RLT (Qiagen) plus β -mercaptoethanol. The bead-bound fraction of RNA and protein was washed 3 $\times$ with cold potassium chloride wash buffer (20 mM RNase-free HEPES potassium hydroxide pH7.4, 5 mM RNase-free magnesium chloride, 350 mM RNase-free potassium chloride, 1% NP-40, 0.5 mM dithiothreitol, 100 μ g/mL cycloheximide in methanol). Beads were then resuspended in RNA lysis buffer RLT (Qiagen) plus β -mercaptoethanol. All samples were then stored at -80°C .

RNA Isolation and Preparation for Quantitative Polymerase Chain Reaction and RNA Sequencing

For quantitative polymerase chain reaction analysis, cDNA was generated using the High-Capacity cDNA Reverse Transcription Kit (4368814; Fisher) per the manufacturer's

instructions. Quantitative polymerase chain reaction reactions were generated using iQ SYBRGreen Supermix (1708880; Bio-Rad) and run on Bio-Rad CFX96. For total intimal-RNAseq, samples were prepared with Illumina TruSeq stranded RNA library and sequenced at 25 million reads per sample, 75 bp single-end reads. For TRAP-RNA sequencing, samples were prepared using Takara Bio v4 SMARTSeq (635026; Takara) and sequenced at 100 million reads per sample, 150 bp paired-end sequencing.

Gene Expression and Alternative Splicing Analysis

FASTQ files were aligned via STAR against the reference genome Gencode M25 (GRCm38.p6), and gene expression was quantified by RSEM. Count tables were then analyzed by DESeq2³² in R. Expression levels were examined by transcripts per million. Intron clustering, requiring 50 split reads supporting each cluster, and differential splice analysis were performed via Leafcutter.³³ Subsequent data analysis to examine pathway enrichment was performed in Jupyter Notebook, with code outlined here (https://github.com/pamurphyUCONN/2024_Nicholas_Elav11).

En Face Immunofluorescence

Mice were perfused through the left ventricle first with cold PBS. Aortas were dissected, visceral fat was removed, and the vessel was flushed again with PBS. Vessels were then pinned to wax and dissected to expose the arterial intimal lining. Vessels were fixed in 0.2% PFA for 1 hour at 4°C, followed by blocking and permeabilization in 1% BSA in PBS containing 0.2% Triton X-100 for 1 hour at room temperature (RT). Vessels were incubated in primary antibody overnight at 4°C in 0.1% BSA with 0.02% Triton X-100. Primary antibodies used were rat anti-mouse VE-Cadherin at 1 µg/mL (550548; BD) and rabbit anti-mouse Elavl1 at 7 µg/mL (200342; Abcam). After 3 washes with cold PBS, vessels were incubated with secondary antibodies for 1 hour at RT in 0.1% BSA with 0.02% Triton X-100. Secondary antibodies used were goat anti-rat AF488 at 2 µg/mL (A11006; Thermo) and biotinylated goat anti-rabbit 2 µg/mL (Jackson Immuno). Vessels were then washed for 5 minutes at RT with PBS, and streptavidin 594 (Thermo) was added to detect biotinylated secondary to Elavl1, along with 1 µg/mL DAPI (4',6-diamidino-2-phenylindole; D9542; Sigma). Two washes with PBS were done before mounting on a slide with Fluoromount (F4680; Sigma) for imaging on the LSM800 confocal microscope with ZenPro software. After image collection with the same instrument settings and during the same session, mean fluorescence intensity was calculated using ImageJ.

Atherosclerotic Plaque Severity Analysis

For both chronic and PCAL models, blood was collected via cardiac puncture through the right ventricle. Mice were then perfused through the left ventricle with cold PBS. Arteries were dissected, adventitia and visceral fat were removed, and the vessel was flushed again with PBS. For chronic hyperlipidemia, plaque in some aortas was directly imaged in most cases by opening the artery lengthwise, pinning it down on black wax, and taking images on a dissection scope with standardized

magnification. Plaque area was quantified using ImageJ by combining all images and thresholding them together to select plaque ROIs.

For partial carotid ligation analysis, the midpoints (containing about $\approx 1/3$ total tissue) from carotid arteries were fixed in 4% PFA overnight at 4°C, then incubated in 30% sucrose in PBS for several hours before embedding in OCT (23730571; Fisher). Carotid arteries were then snap-frozen and stored at -80°C until cryo-sectioning at 10 µm. Oil Red O staining was performed by incubating tissues with a 0.22 filtered 3:2 dilution of 6.25 g/L Oil Red O in isopropanol to distilled water for 5 minutes before washing with tap water. Slides were mounted and then imaged. Percent occlusion was quantified using ImageJ.

Plaque Immunophenotyping

Vessel and Lymph Node Digestion

Aortic vessels were obtained after PBS perfusion, and visceral fat and adventitia were removed. Vessels were mechanically digested via mincing for 4 minutes each and then enzymatically digested for 1 hour at 37°C with gentle rotation. Mesenteric and cervical lymph nodes were isolated and minced similarly, before enzymatic digestion. Enzymatic digestion cocktail was prepared in balanced salt solution with 150 U/mL collagenase type IV (C5138; Sigma-Aldrich), 60 U/mL DNase I (Sigma-Aldrich), 1 µmol/L MgCl_2 (Sigma-Aldrich), 1 µmol/L CaCl_2 (Sigma-Aldrich), and 5% fetal bovine serum (FBS). Digested tissue was gently crushed over 35 µm nylon mesh cell filter caps and digestion was quenched with 10% FBS in balanced salt solution. Samples were centrifuged for 5 minutes at 300×g at 4°C, supernatant was aspirated, and the cell pellet was resuspended in flow cytometry buffer (fluorescence-activated cell sorter [FACS] buffer, 2% FBS, and 1 mmol/L EDTA in PBS).

Blood Collection

Blood was collected via cardiac puncture at the time of collection. An aliquot was reserved for serum cholesterol analysis. The remainder was immediately added to 0.55 g/L heparin (H3393-50KU; Sigma) in HBSS and temporarily stored on ice. Samples were centrifuged for 5 minutes at 300×g. Supernatant was aspirated, and pellets were resuspended in Cold Spring Harbor Red Blood Cell Lysis (155 mmol/L NH_4Cl , 12 mmol/L NaHCO_3 , 0.1 mmol/L EDTA) and incubated at RT for 10 minutes, then quenched with PBS. Lysis was repeated if needed. Samples were centrifuged for 5 minutes at 300×g at 4°C, the supernatant was aspirated, and the cell pellet was resuspended in FACS buffer.

Immunostaining

Cell pellets were resuspended in the following antibody cocktail diluted in FACS buffer: 1 µL per test Fc Block (InVivoMAb anti-mouse CD16/32, BioXcell BE0307), LIVE/DEAD UV Blue (1:100, Thermo Fisher, L34962), CD8 (1:200, 100708, clone 53-6.7; Biolegend), CD4 (1:200, 100536, clone RM4-5; Biolegend), CD45.2 (1:200, 103116, clone 30-F1111; Biolegend), Cd3e (1:200, 100348, clone 145-2C11; Biolegend), Gr1 (1:200, 552093, clone RB6-8C5; B.D. Pharmingen), and Cd11b (1:200, 101212, clone M1/70; Biolegend). Full-minus-one controls were used to set gates. LSRIIA (BD Biosciences) was used for acquisition. All flow cytometry data were analyzed with FlowJo (Tree Star, Ashland, OR).

10x Single-Cell Cellular Indexing of Transcriptomes and Epitopes by Sequencing for Chronic Hyperlipidemic Samples

Whole and PBS perfused aortic tissues were similarly dissected of adventitia, followed by a slightly modified digestion protocol. Vessels were minced for 4 minutes per sample and then enzymatically digested at 37°C at 265 RPM for 50 minutes. Digestion cocktail was prepared in HBSS (24020117; Fisher) with CLS-1 (collagenase type I; Worthington) 1000 U/mL, collagenase type XI (C7657; Sigma) 180 U/mL, hyaluronidase type I-S (H3506; Sigma) 80 U/mL, Dnase I (DN25; Sigma) 80 U/mL, and FBS (1:200). Digestion was quenched with 100% FBS before tissue was gently crushed over 100 µm nylon mesh cell filters. Samples were centrifuged for 5 minutes at 300×g at 4°C, the supernatant was aspirated, and the cell pellet was resuspended in FACS buffer.

Aortic samples were stained for flow cytometry in FACS Buffer with 1 µL per test Fc Block (InVivoMAb anti-mouse CD16/32, BioXcell BE0307), anti-mouse CD45 (1:200, 146603, clone I3/2.3; Biolegend), CD31 (1:200, 102509, clone MEC13.3; Biolegend), and ICAM2 (1:200, A15451, clone 3C4; Invitrogen). To hashtag samples for 10x reaction, cells from each tissue were stained with a unique TotalSeqB anti-mouse hashtag (1:200, Biolegend) and incubated for 30 minutes at 4°C. DAPI was added to aortic samples immediately before FACS to distinguish live cells. CD45+ live immune cells from all aortic samples were collected into 1 tube, and CD45-CD31+ICAM2+ live ECs from all aortic samples were collected into another protein-lo-bind 1.5 mL Eppendorf tube, precoated with 1% BSA in PBS.

All FACS-sorted cells were then stained with cellular indexing of transcriptomes and epitopes by sequencing TotalSeqB mouse Universal Cocktail (199902; Biolegend) and TotalSeq-B0924 CD257 OX40 L (108825; Biolegend) for 30 minutes at 4°C for cellular indexing of transcriptomes and epitopes by sequencing analysis. An equal number of white blood cells from each sample was combined and then also stained with this antibody cocktail under the same conditions. After washing, ~40 000 cells from blood and 40 000 cells from aorta were then combined into a single 10x reaction (Chromium Next GEM Single Cell 3' Kit v3.1) and run on a Chromium system to generate a single-cell cDNA library. Steps followed the 10x protocol, with the addition of a 3-minute extension during library amplification to reduce amplification bias.

Analysis of 10x Single-Cell Cellular Indexing of Transcriptomes and Epitopes by Sequencing

Cellranger (7.0.1) was used to map reads against Mouse (mm10) 2020-A and to generate gene-level and antibody-level read counts. Paths to the feature library (antibody capture and gene expression) were specified during the processing, so we ended with an h5 object that included both gene expression and antibody capture information. Raw h5 feature-barcode matrix files were analyzed using the Seurat package (5.0.3). Hashed samples were demultiplexed using the MULTISEQDemux function (autoThresh=TRUE), which is part of the deMULTiplex R package, and cells labeled with multiple, or no hashtags were removed. Cells with <300 or >5000 genes, >20 000 UMI, or >10% mitochondrial reads were filtered out. Remaining doublets were filtered out using scDblFinder Bioconductor package (1.18.0). Data from all objects was integrated and

normalized using the sctransform R package (0.4.1), and principal component analysis was performed using the 3000 most variable genes and "RunPCA" function in the Seurat Package. Clustering was carried out using the first 20 PCs and the "RunUMAP," "FindNeighbors," and FindClusters functions in the Seurat package. UMAPs were visualized using the "DimPlot" in the Seurat package. Clusters were annotated manually by analyzing enriched transcripts. Subclustering was performed by subsetting clusters associated with cell types of interest and performing data integration (sctransform) and UMAP clustering as described above. Differential expression analysis was performed by using the "FindMarkers" function in the Seurat package with the Wilcoxon rank-sum test algorithm, and differences with and adjusted $P < 0.05$ were considered significant. Antibody barcode counts were normalized using centered log ratio normalization, and differential expression of normalized antibody counts was performed using the "FindMarkers" function in Seurat as previously mentioned.

Generation of In Vitro EC Line

Mouse aortic ECs were isolated from *RosaCreER; Elavl1f/f* adult male mice as previously described.²⁰ Briefly, the vasculature was perfused with cold PBS through the left ventricle. The descending aorta was dissected and filled with 2% collagenase type II (LS004176; Worthington) in serum-free DMEM, then closed at the ends with 7-0 suture. The digesting vessel was incubated for 45 minutes at 37°C in DMEM+20% FBS. Subsequently, digested intimal cells were flushed out with vascular basal medium+EC growth kit VEGF (vascular endothelial growth factor; ATCC, PCS-100-030 and PCS-100-041). Flushed cells were centrifuged at 300×g for 3 minutes, supernatant aspirated, and cells were plated on rat tail collagen type I (A1048301; Fisher) in EC growth medium (as above) supplemented with primocin (1:500, ant-pm-1; Invivogen). Intimal cells were expanded for 48 hours and then treated with lentiviral TetOn-SV40 and polybrene (1:1000, TR-1003-G; Sigma). The viral vector was constructed as described.²⁰ Cells were expanded in EC growth medium and 2 µg/mL doxycycline (D9891-1G; Sigma). For FACS purification, cells were stained for 30 minutes at 4°C in FACS buffer with CD45-antigen-presenting cell (APC)-Cy7 (Clone 30-F11, 103116; Biolegend) and ICAM2-PE (Clone 3C4, A15451; Thermo) and FACS sorted (BD ARIAII) for single-cell CD45-ICAM2+. Sorted aortic ECs were expanded in the presence of doxycycline, immortalized with retroviral pBABE-neo-hTERT (1774; Addgene), and then allowed to expand. To induce excision of *Elavl1* in *RosaCreER; Elavl1f/f* immortalized mouse aortic ECs before coculture experiments, cells were treated with a final concentration of 1 µM 4-hydroxytamoxifen, active tamoxifen metabolite(Z; H7904; Sigma) for 3 days or with 100% EtOH daily as a vehicle control.

Western Blot

Media was collected, and cells were washed 3× with PBS. Cell lysates were collected in RIPA buffer using cell scrapers and then stored at -20°C. Lysate was thawed, treated with 1 mol/L dithiothreitol (A39255; Thermo), incubated at 95°C for at least 5 minutes, and then diluted 1:1 with 2× Laemmli buffer. Samples were run on Novex WedgeWell 4% to 20% Tris-Glycine Mini Gel (XP00100BOX; Thermo) at 125 V for

1.5 hours. Transfer to membrane was done using Trans-Blot Turbo Transfer System (1704150; Bio-Rad) and transfer kit (1704273; Bio-Rad).

Membranes were first blocked in 5% milk in tris buffered saline+0.1% Tween-20 for 30 minutes at RT. Membranes were then incubated in primary antibody for Elavl1 (1:1000, Abcam, 200342) or C1q (1:500, Abcam, ab182451) ON at 4°C in 0.5% milk in Tris buffered saline+0.1% Tween-20. After washing in Tris buffered saline+0.1% Tween-20, membranes were incubated in secondary antibody for anti-rabbit-HRP (1:5000, 111-035-144; Jackson Immuno) for 1 hour at RT. After developing and imaging with Clarity ECL Western blot substrate (1705060; Bio-Rad), the membrane was washed again. The membrane was then blocked in 5% milk Tris buffered saline+0.1% Tween-20 for 30 minutes at RT, incubated with BetaActin-HRP (1:1000, Invitrogen, MA5-15739-HRP) for 3 hours at RT. Washing, developing, and imaging followed staining as described above. Quantification was performed using ImageJ.

CD8 Staining of Tissue Sections

Zinc-fixed paraffin-embedded innominate artery sections of 9-month old EC knockout and WT control mice were stained alongside an additional WT mouse lymph node positive control. First, the slides are heat deparaffinized in an oven at 60°C for 30 minutes, followed by a chemical deparaffinization; histoclear 10 minutes, histoclear 10 minutes, 100% ethanol 5 minutes, 100% ethanol 2 minutes, 95% ethanol 2 minutes, 70% ethanol 2 minutes, MiliQ water 5 minutes. Next, the tissue sections are circled with a pap pen. The tissue is then treated with Sudan Black (0.5% in 70% ethanol, 22 µm filtered), incubated at RT for 30 minutes, adding more as necessary so they do not dry out. After incubation, the tissue was washed 3× with 1× PBS for 5 minutes, and the tissue was blocked in a solution of 5% normal goat serum and 0.1% Triton X-100 in 1× PBS for 30 minutes at RT. Primaries are applied in a buffer of 0.5% normal goat serum and 0.01% Triton X-100 in 1× PBS and incubated overnight at 4°C (rtCD8a Invitrogen [14-0808-82] at 1:100 or rtIgG isotype control at 1:400). Following another three 5-minute washes in 1× PBS, the tissue is incubated in secondaries in 0.5% normal goat serum and 0.01% Triton X-100 in 1× PBS for 2 hours at RT (α rtbiotin). After washing 3 more times with 1× PBS for 5 minutes, the tissue is incubated in tertiaries (DAPI at 1:2000 and SA 647 at 1:2000) in 0.5% normal goat serum and 0.01% Triton X-100 in 1× PBS for 2 hours at RT (DAPI 1:2000, SA 647 1:2000). Finally, the tissue is washed 3× with 1× PBS for 5 minutes, mounted with a coverslip using Fluoromount, and stored at 4°C for imaging with Alexa 647, Alexa 488, and DAPI.

In Vitro Coculture System

APC Generation

Bone marrow was collected from the femurs and tibia of adult male C57BL/6J mice. Briefly, bones were dissected, ends removed, and placed open end down into 0.5 mL Eppendorf tube with 18G needle puncture. The bones and punctured 0.5 mL tube were inserted into a 1.5 mL Eppendorf and then centrifuged at 21 200×g for 15 to 20 seconds. Pelleted bone marrow was resuspended in PBS, passed over a 100 µm filter, and centrifuged again at 300×g for 5 minutes. Supernatant was

decanted, and cells were treated with red blood cell lysis buffer (155 mM ammonium chloride, 12 mmol/L sodium bicarbonate, 0.1 mmol/L EDTA) for 1 minute. Red blood cell lysis was quenched with FACS buffer. Cells were pelleted by centrifugation at 300×g for 5 minutes, then counted on Luna cell counter (L20001; New England BioGroup). Monocytes were then isolated using EasySep Monocyte Isolation Kit (19861; StemCell Tech). Monocytes were seeded in immune cell medium (MEM with L-glutamine, 10% FBS+50 mL Tumor Cocktail, 2 µg/mL primocin) supplemented with 20 ng/mL GM-CSF (granulocyte-macrophage colony-stimulating factor, 576304; Biolegend) on nontissue culture-treated plates. After 3 days of culture, cells were fed with fresh media and 40 ng/mL GM-CSF. On day 5, nonadherent and loosely adherent cells were collected, resuspended in APC cryo preservation media (90% immune cell media, 10% dimethyl sulfoxide, 20 ng/mL GM-CSF), and stored at −80°C.

CD8⁺ T-Cell Isolation

Spleens from OT-I *Rag*^{−/−} transgenic mice were dissected and then crushed over a 100 µm cell strainer. Cells were pelleted by centrifugation at 300×g for 5 minutes. Supernatant was decanted, and pellets were resuspended in red blood cell lysis buffer in 1 to 2 mL for 2 to 3 minutes. Lysis was quenched with PBS or balanced salt solution. Lysed cells were then passed over a 70 µm cell strainer, pelleted by centrifugation at 300×g for 5 minutes, then counted on a Luna cell counter. CD8 T cells were isolated using EasySep CD8 T-cell isolation kit (19853; StemCell). Isolated CD8 T cells were pelleted and then resuspended in 90% FBS+10% dimethyl sulfoxide for cryopreservation and stored at −80°C.

Coculture Experimental Design

The *RosaCreER; Elavl1f/f* EC line (+TetOn-Sv40 and +hTert) was plated in 96-well plates at 1×10⁴ cells per well (day 4), then treated with 1 µM 4-hydroxytamoxifen or EtOH daily for 3 days, starting on the day of plating (D-4 to D-2). The next day (D-1), 4.4×10³ thawed APCs were added per well to EC culture, along with or without antigen, either 0.5 µg/mL EndoGrade OVA (321000; BioVender) or 1 µg/mL SIINFEKL (processed peptide, antigen for OT-I cells; OVA 257-264; S7951; Sigma-Aldrich). The following day, fresh media without antigen was changed for wells treated with SIINFEKL. Subsequently, 2.1×10⁴ thawed naive CD8 OT-I T cells were added to all wells regardless of antigen treatment. For the EdU incorporation assay, 1×10⁵ thawed naive CD8 T cells were added per well along with 1 µM EdU (A10044; Thermo). For cocultures collected at D13, 100 µL supernatant was removed, and 100 µL of fresh media with or without antigen was replaced.

Cocultured supernatant and cells were then collected 3 days or 13 days later for assay by western or flow cytometry. At the final collection point, supernatant was collected and stored at −20°C or −80°C until use. For flow cytometry, nonadherent cells (T cells and the majority of APCs) were first removed from all wells. Then adherent cells (ECs and a few APCs) were collected via 0.25% Trypsin in Versene for 5 to 10 minutes at 37°C. Trypsin was quenched with FBS, and cells were then processed for flow cytometry.

Flow Cytometry Staining

CD45.1⁺ OT-I cocultured cells were stained with an anti-body cocktail diluted in FACS buffer: 1 µL per test Fc Block

(InVivoMAb anti-mouse CD16/32, BioXcell BE0307), LIVE/DEAD UV Blue (1:100, L34962; Thermo Fisher), CD45.1 (1:200, 25-0453-82, clone A20; Thermo), CD45.2 (1:200, 109806, clone 104; Biolegend), CD8 (1:200, 100712, clone 53-6.7; Biolegend), CD44 (1:200, 63-0441-82, clone IM7; Invitrogen), and ICAM2 (1:200, A15451, clone 3C4; Invitrogen). Bio-Rad ZE5 was used for acquisition. All flow cytometry data were analyzed with FlowJo (Tree Star, Ashland, OR). Flow cytometry results were analyzed for CD8 T-cell count, viability, and proliferation, along with APC count and viability, and EC count and viability. One biological experimental replicate of D3 experiments in the study was excluded based on the lack of APCs recovered at the time of collection in the full-length OVA-treated wells. Biological experimental replicates of D13 collection were excluded based on the following criteria: reduced APC viability across all conditions compared with the majority of experimental replicates; statistical outliers, and use of female CD8 T cells as compared with male T cells. The replicate using female T cells had a trend towards fewer T cells detected in the no antigen condition as compared with the other replicates. ECs and APCs were both derived from male mice.

Adoptive Transfer for Analysis of Recruitment to Low-Flow Carotid Artery

CD8 OT-I T cells were isolated as above. Elavl1 ECKO and WT mice underwent PCAL and induction of hyperlipidemia. Two weeks later, 2×10^5 naïve CD8 OT-I cells were intravenously transferred into PCAL recipient mice. Within 24 hours of transfer, recipient mice received intraperitoneal injection of 50 μ g SIINFEK²⁵⁷⁻²⁶⁴ peptide (S7951-1MG; Sigma), 20 μ g of agonist anti-OX40 (clone OX-86 mAb, BE0031; BioXcell), and 10 μ g of agonist anti-41BB (clone 3H3 mAb, BE0239-1MG; BioXcell). One week posttransfer, carotid arteries were collected, digested as described above, and assayed via flow cytometry. Digested cells were stained with the following antibody cocktail diluted in FACS buffer: 1 μ L per test Fc Block (InVivoMAb anti-mouse CD16/32, BioXcell BE0307), Live/Dead Zombie Aqua (1:100, 423101; Biolegend), CD45.1 (1:200, 25-0453-82, clone A20; Thermo), CD45.2 (1:200, 109823, clone 104; Biolegend), CD3e (1:200, 100348, clone 145-2C11; Biolegend), CD4 (1:200, 100536, clone RM4-5; Biolegend), CD8 (1:200, 100708, clone 53-6.7; Biolegend), and Va2 (1:200, 46-5812-80, clone B20.1; Invitrogen) and Vb5 (1:200, 139503, clone MR9-4; Biolegend). LSRIIA (BD Biosciences) was used for acquisition. All flow cytometry data were analyzed with FlowJo (Tree Star, Ashland, OR).

Statistical Analysis

With the exception of single-cell analysis (statistical comparisons performed in Scanpy), statistical analysis was performed in GraphPad Prism. For paired samples with equal variance, a Student *t* test was performed, a Mann-Whitney *U* test was performed. For comparisons among multiple samples, we used ANOVA with post hoc Sidak multiple comparisons test when variances were equal and Kruskal-Wallis with post hoc Dunn when they were not.

To assess the influence of biological sex, data were initially analyzed using a 2-way ANOVA (genotype $\times$ sex). In instances where no significant interaction or main effect of sex was

observed ($P > 0.05$), male and female data were pooled for the final analysis to increase statistical power. If a significant sex-dependent effect was detected, data were analyzed and presented separately by sex. For key metrics, including plaque phenotype and CD8+T-cell recruitment, sex-disaggregated data are provided in the Supplemental Information even in the absence of significant sex-dependent effects.

RESULTS

Ribosomal Profiling of ECs in Atherosclerotic Lesions Identifies Changes in Splicing and Translation of Immune-Regulatory Pathways

To examine the transcriptional changes in EC within well-developed atherosclerotic plaque, we employed a model of low flow induced by PCAL coupled with hyperlipidemia mediated by adenoviral delivery of mutated PCSK9 (proprotein convertase subtilisin/kexin type 9; AAV-mPCSK9) and high-fat diet.³⁴ Surgical ligation induces low and disturbed flow along the length of the left carotid artery while the contralateral right carotid artery continues to experience normal flow, and thus serves as a valuable animal-specific contralateral control (Figure 1A). Previously, this model has been used to assess short-term EC responses to disturbed flow by rapid intimal mRNA isolation using a Trizol flush method.^{7,20,22} Significant plaque development occurs by 3 weeks,³⁴ at which point there is substantial interaction between the immune system and ECs. Therefore, we chose to examine transcriptional alterations in arterial ECs 3 weeks after the induction of disturbed flow.

To assess mRNA translation, we used TRAP for the EC-specific isolation of ribosomes and associated RNA in situ.²⁷ Inducible VE-Cadherin-Cre activity (*Cdh5-CreERT2; RiboTRAP*) results in the selective expression of a GFP-tagged L10a ribosomal protein in ECs, which can be immunoprecipitated with associated RNA. Typically, this is done from a whole tissue homogenate, but we adapted this protocol based on our experience with rapid intimal Trizol flush to quickly isolate highly purified ribosomal complexes from the arterial intima (Figure 1B and 1C). This flush substantially increases the specificity of the endothelial mRNA isolation above whole tissue homogenate by providing 2 levels of purification. We first validated the flushing of mRNA by examining EC transcript CD31/Pecam in the flush and vessel remnant from aortic and carotid tissue, and found the majority of transcript in the flush, similar to Trizol extraction (Figure S1B). We then applied this approach to more complex atherosclerotic lesions, where Trizol flush of total intima mRNA (total in Figure 1) is typically contaminated by a low level of smooth muscle and myeloid transcripts from the neointima. We found that, while the TRAP approach approximated *Icam2* and *KLF2* levels of Total mRNA isolation in parallel experiments, and the suppression of

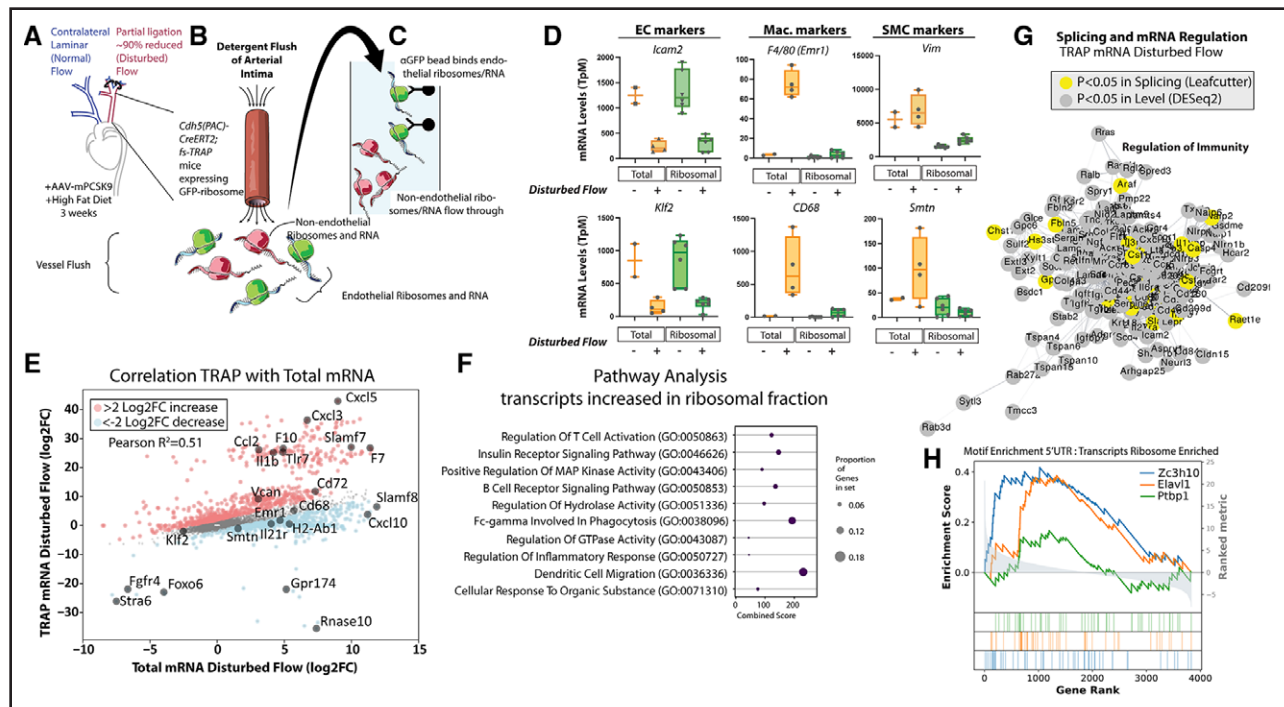


Figure 1. Translating ribosome affinity purification (TRAP) from atherogenic endothelium identifies changes in immune-regulatory pathways.

A, Partial carotid arterial ligation (PCAL) induction of low-flow in the partially ligated left carotid artery and hyperlipidemia collected at 3 weeks post-PCAL. **B**, Approach for isolating endothelial-specific ribosomal mRNA from both low-flow and normal flow vessels via RiboTRAP, followed by **(C)** magnetic bead enrichment of eGFP-labeled ribosomes. **D**, Expression level (FPKM) of indicated genes in disturbed flow and normal flow vessels, showing levels in TRIzol-mediated extraction vs TRAP isolation. **E**, Plot showing the effects on transcript levels between low and disturbed flow and contralateral normal flow arteries, between TRAP (y axis) and total mRNA (x axis). **F**, Gene set enrichment analysis of low and disturbed flow-regulated transcripts thresholded on transcripts per million (TPM) ≥ 100 . Top 300 genes up indicate the genes and associated transcripts most enriched in TRAP low vs contralateral relative to Total mRNA low vs contralateral (red in **E**). **G**, Cytoscape visualization of a top Gene Ontology (GO) category (regulation of immunity) associated with disturbed flow-regulated transcripts in ribosomal mRNA. Analysis represents interactions in StringDB, in which each node is a gene, and connections are represented by strings. There is no significance of distance between nodes. **H**, Enrichment of Cis-RNAbp motifs in the 5' untranslated region (UTR) of transcripts enriched at the ribosome (red in **E**) vs those disenriched (blue in **E**), showing the relative enrichment for Zc3h10, Elavl1, and Ptbp1 6mers. **A** through **H**, Total intimal-RNAseq by Trizol extraction (total; $n=4$ wild-type [WT] ligated vs $n=2$ contralateral, 2 male and 2 female, and 1 male and 1 female) compared with TRAP RNAseq (TRAP; $n=6$ WT ligated vs $n=6$ Contralateral, 3 male and 3 female in each; **G**) gray=differential mRNA TRAP, DESeq2 P adjusted < 0.05 , yellow=differential mRNA splicing Leafcutter P adjusted < 0.05 . EC indicates endothelial cell; GFP, green fluorescent protein; and SMC, smooth muscle cell.

these transcripts in the low-flow artery, that macrophage enriched transcripts were reduced, as were smooth muscle cell-associated transcripts in the TRAP fraction (Figure 1D). Moreover, looking at bead enrichment within the vessel flush (already $>99\%$ endothelial transcript²⁰), we found further enrichment of endothelial transcript (Figure S1C). Thus, TRAP effectively enriched EC transcripts and provides new insight into the ribosomal association of transcripts in atherosclerosis.

In general, as indicated by expression of KLF2 (Figure 1D) and correlational analysis (Figure 1E; $R^2=0.51$), expression changes in the ribosome-associated mRNA resembled those in total mRNA from Trizol flush. However, we noted that the responses did not tightly correlate, and that some transcripts were relatively unchanged by the flow response in total mRNA, but strongly changed in their ribosomal enrichment (eg, *Cxcl5*, *Cxcl3*, *F10*, *Il1b*; Table S1; Ribo_v_Total_Diff).

Pathway analysis of these transcripts indicates regulation of immune responses in both groups (Figure 1F; Figure S2). An examination of the overall transcriptional responses in the low-flow atherosclerotic endothelium in Total and TRAP mRNA indicated similar pathways, notably also containing T-cell activation (Figure S3). We also examined mRNA splicing changes in the ribosomal fraction, identifying >3000 significant changes by Leafcutter splicing analysis (Figure S4A; Table S2; Splice_Ribo_Lig_v_Contra). One example is a $\approx 40\%$ increase in the inclusion of a skipped exon in *Irf2*, a critical regulator of MHC-I (major histocompatibility complex type I) activity, and CD8 interactions and immune evasion in parenchymal cells³⁵ (Figure S4B). Pathways affected by changes in both TRAP-associated mRNA levels and alternative splicing were enriched for immune processes (Figure 1G; Figure S5). For example, transcript levels and alternative splicing patterns

associated with Regulation of Immunity were found to be regulated by low flow in the TRAP data set (Figure 1G). Thus, TRAP reveals changes in the ribosome-associated mRNA occurring in the endothelium in atherosclerotic plaque of hypercholesterolemic mice.

Given the large changes in ribosome-bound transcripts, we wondered whether differentially recruited transcripts were enriched for particular RBP motifs in the 5'UTR region associated with translational regulation. To examine this, we obtained 5'UTR regions from Gencode data for transcripts preferentially bound (red in Figure 1E) or reduced in binding (blue in Figure 1E). We then examined 6-mer RBP motifs enriched in red versus blue UTR regions. Finally, with this list of enriched UTR regions, we looked for RBP binding sites, defined by *in vitro* binding assays (Cis-BP RNA data set).³⁶ The most highly enriched binding sites in the red UTR regions were Rbm4b, Cpeb3 and 4, Pcbp1, 2, 3, and 4, Cnot4, Tial1, and Tia1 and Elavl1 (Figure 1H; Figure S6). Cpeb proteins are primarily known for translational control,³⁷ and the other proteins also have established roles in mRNA translation regulation. Of these proteins, Elavl1 also appeared at the top of the list in a similar analysis of the 200 nt upstream and downstream of mRNA splicing events induced at the ribosome under altered flow (Figure S4C), indicating that this RBP was likely to be important in both translation and splicing in the endothelial response to disturbed flow.

Thus, TRAP data identifies ribosome enrichment of transcripts and splice isoforms predicted to affect immune-endothelial interactions, and motif analysis points to Elavl1 as 1 key regulator.

RBP Elavl1 Determines the Splicing and Ribosomal Association of Immune-Regulatory Pathways in Plaque Endothelium

To examine Elavl1 protein levels in the endothelium in sites of disturbed flow, we performed immunostaining of disturbed flow regions of the inner curvature of the aortic arch and laminar flow regions of the descending aorta (Figure 2A). We found that Elavl1 protein was increased in ECs of the arch when compared with nondisturbed flow sites in the descending aorta (Figure 2B and 2C). To genetically define the role of EC Elavl1, we combined existing alleles to generate a novel, temporally inducible, EC-specific Elavl1 knockout (ECKO) mouse model (*Cdh5-CreERT2; Elavl1^{fl/fl}*). Cre recombinase expression driven by VE-Cadherin (*Cdh5*) results in EC-specific excision of a region of Elavl1 containing the translation start site. Tamoxifen-induced recombination of mice at least 6 weeks old resulted in a reduction of Elavl1 protein staining (Figure 2B and 2C) and reduced levels of the excised region in transcripts from ECKO intimal RNA (*Cdh5-CreERT2; Elavl1^{fl/fl}*) relative to WT littermate controls (*Elavl1^{fl/fl}*; Figure S7).

We isolated mRNA from ECKO mice in the PCAL+AAV-PCSK9 model with either total mRNA flush or TRAP isolation. Overall, total mRNA expression changes of atherogenic endothelium assessed by DESeq2 were comparable in ECKO mice and controls (Figure 2D; Pearson correlation, 0.91). Although Elavl1 can regulate mRNA stability,³⁸ loss of Elavl1 did not substantially affect total mRNA levels or the levels of canonically stabilized transcripts in this model (Figure 2D; Figure S8). This was true even for well-known Elavl1 stabilized transcripts (Figure S7). However, differences occurred at the level of ribosome association, where the response of Elavl1 ECKO mice diverged from WT mice (Figure 2E; Pearson correlation, 0.6). This led to many more differences appearing between ECKO and WT mice when looking at the ribosomal fraction versus total mRNA (138 versus 1, at *P* adjusted <0.05, Table S1). Analysis of one such ribosome-associated transcript, *C1qa*, shows an increase in low-flow conditions, as previously reported,³⁹ which is further increased with the loss of Elavl1 (Figure 2F). Thus, Elavl1 deletion affects ribosomal association of mRNA, both at baseline and under disturbed flow conditions.

To examine pathways in the transcripts regulated by Elavl1 at the ribosome, we performed ranked gene set enrichment analysis of the DESeq2-defined changes in transcript levels in TRAP mRNA (Figure 2G). We did a similar analysis of the changes in mRNA levels in the total intimal flush from mice performed in parallel. This revealed suppression of transcripts associated with mitochondrial respiration and translation (Figure 2G), and an enrichment of genes modulating the immune response and TCR signaling (Figure 2G). An example gene set enrichment analysis plot shows enriched terms for T-cell activation pathways (inset from Figure 2G). Notably, these effects appeared in the normal flow contralateral artery as well as the low-flow ligated artery, although they were generally enhanced under low and disturbed flow. Generally, the effect of Elavl1 depletion was to enhance ribosomal changes already occurring in the endothelium under these chronic low-flow conditions (Figure S9A), enhancing the ribosomal association of transcripts regulating immune and complement responses and reducing transcripts associated with clotting functions and angiogenesis (Figure S9B).

We used Leafcutter to examine the change in mRNA splicing in total intimal RNA or TRAP isolated mRNA from the same study. We found 355 changes in mRNA splicing in TRAP mRNA (*P* adjusted <0.05), but most did not overlap with transcripts changing in levels in the TRAP mRNA—even when loosening restrictions to an unadjusted *P* value of 0.05 (Figure S10; Table S2). However, similar pathways were affected by both ribosomal total levels of expression and by splicing of transcripts at the ribosome (Figure S11). Although most transcripts were affected by either expression or splicing, an exception

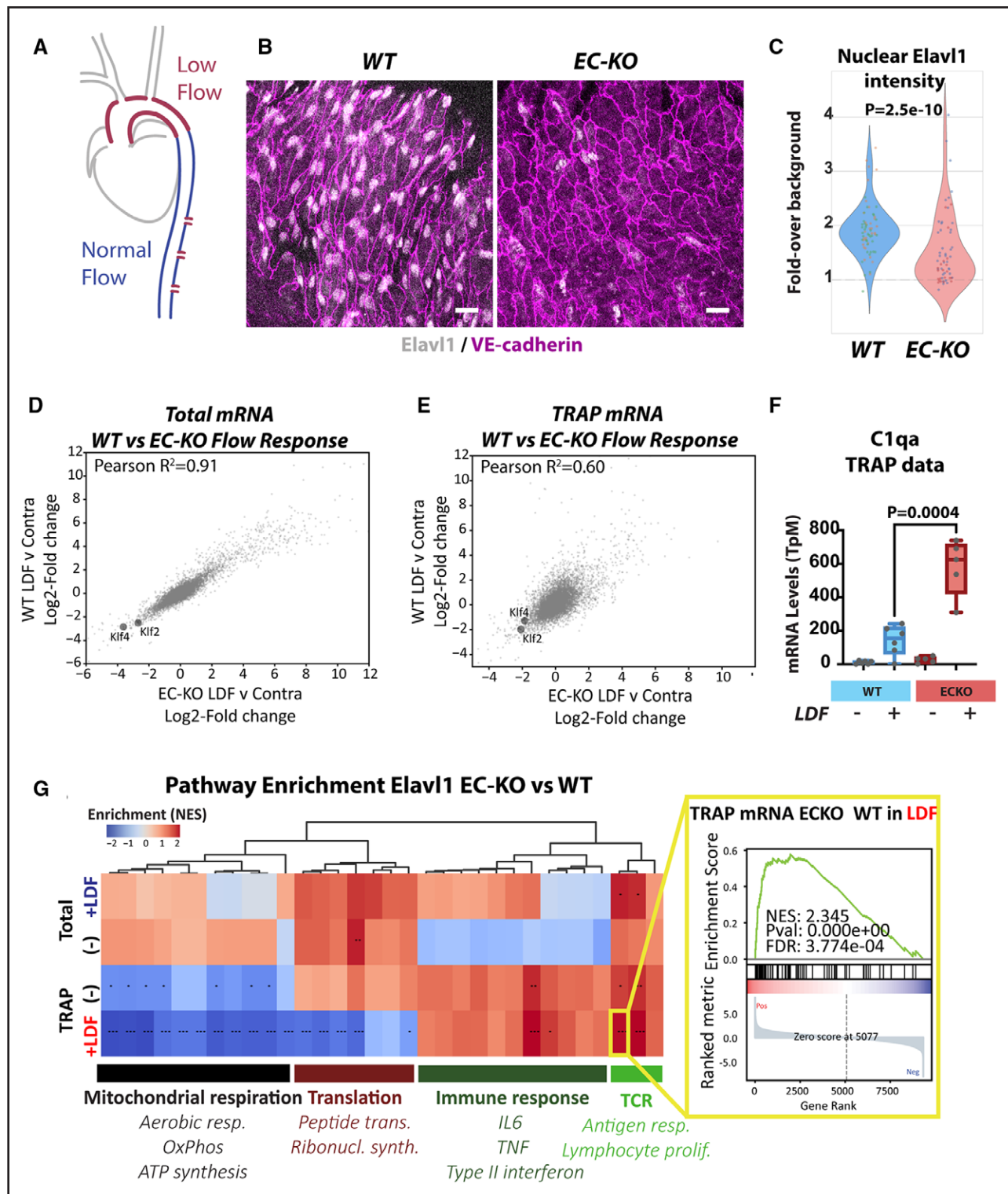


Figure 2. Endothelial Elavl1 is increased in atherogenic endothelium and regulates ribosome-associated mRNA.

A, Schematic showing regions of the inner aortic arch exposed to disturbed flow and the descending aorta exposed to laminar flow. **B**, Elavl1 immunostaining in nuclei of aortic arch endothelial cells prepared en face. **C**, Elavl1 mean fluorescence intensity for each nucleus, as fold increase over background. Each dot represents 1 nucleus, data represents 4 images per animal, 2 animals, male and female for each genotype. **D** and **E**, Plots showing correlations in gene expression between wild-type (WT) and postnatal endothelial-specific Elavl1/HuR deletion (ECKO) carotid intima in the partial carotid artery ligation model, when total Trizol flushed mRNA is assessed (**D**) or translating ribosome affinity purification (TRAP)–mRNA is assessed (**E**). Data shown are presented as log2-fold change in expression over contralateral artery. $n=4$ ligated and $n=2$ contralateral for WT Total RNA, $n=4$ ligated and $n=2$ contralateral for ECKO Total RNA. $n=6$ ligated and $n=6$ contralateral for WT TRAP RNA and $n=5$ ligated and $n=5$ contralateral for ECKO TRAP RNA. **F**, Example plots showing the mean expression level for (Continued)

was CD27 (Figure S12), an important costimulatory molecule, which showed increased levels of expression and increased splicing, leading to an exon 3-inclusive transcript in ECKO TRAP mRNA.

Thus, deletion of Elavl1 affects both splicing of mRNA transcripts and their association with the ribosome under low-flow conditions, enhancing an increase in ribosomal transcript levels already occurring to some extent in chronic low-flow conditions.

Endothelium-Specific Deletion of Elavl1 Reduced CD8 T Cells and Activation in Plaque

To specifically examine the effects of EC Elavl1 deletion on endothelial interactions with immune cells in plaque, we tested 2 mouse models of atherosclerosis (Figures 3 and 4). Both models developed plaque at sites of disturbed flow, specifically in the partially ligated left carotid artery 3 weeks after PCAL and hypercholesterolemia (Figure 3B) or at natural sites of disturbed flow in the aorta after 3 to 9 months of chronic hypercholesterolemia (Figure 4B). We observed decreased levels of plaque in both models, with a mean carotid occlusion reduced from $\approx 50\%$ in WT to $\approx 10\%$ in ECKO mice in the PCAL model (Figure 3C; $P < 0.05$), and reduced vessel plaque coverage in descending aortas of male mice from $\approx 50\%$ in WT to $\approx 20\%$ in ECKO after 9 months of chronic hypercholesterolemia (Figure 4C; $P < 0.05$). This occurred despite similarly increased cholesterol levels in WT and ECKO mice (Figure 3A; Figure S13). Notably, deletion of Elavl1 affected lesion progression, but not initial development. By 3 months, plaque development in both WT and ECKO mice was minimal and comparable, with only a few initial loci visible in the aortic arch and descending aorta (Figure 4C). Thus, although both WT and ECKO mice initially developed similar levels of plaque, growth was slower in ECKO mice.

Given the prominent effects of Elavl1 deletion on transcriptional responses related to the immune response and T-cell receptor signaling, we reasoned that T-cell levels in atherosclerotic lesions might be impacted by Elavl1 expression. Murine ECs constitutively express MHC class I but not MHC class II molecules, allowing direct interactions with CD8 but not CD4 T cells.^{40,41} Furthermore, CD8 T cells consistently associate with plaque development, so we chose to focus on them.^{17,19} By generating unilateral disturbed flow and subsequent atherosclerosis, the PCAL model provides an animal-specific control for studying immune cell recruitment. In line with

prior studies, we observed a 3-fold increase in CD8+T cells in the ligated artery versus the contralateral control (Figure 3D and 3E).^{18,42} In contrast, ECKO mice failed to show this expected increase under low and disturbed flow conditions (Figure 3D and 3E; $P < 0.01$). This reduction in CD8 T-cell accumulation was similar across male and female ECKO mice (Figure S14), and notably did not always coincide with substantial reductions in other immune cell subsets in plaque (CD45 and CD11b; Figure 3D and 3E). Similarly, this difference in plaque CD8 T cells does not reflect a lack of circulating cells, as there were no differences in CD8 T cells in either blood or lymph node (Figure S14). Thus, CD8 T-cell accumulation in plaque was strongly suppressed by the deletion of endothelial Elavl1 in the PCAL model.

Next, we tested whether CD8 accumulation was dictated by plaque size or if it was independent of plaque development. To address this, we examined aortic arches of hypercholesterolemic mice at 3 months of age, where we found plaque size was not substantially affected by ECKO (Figure 4C). We measured both the total counts of CD8 T cells, and their ratio to CD45+ cells in males and females. There was a $\approx 70\%$ reduction in CD8 T-cell numbers in both males and females. To determine if this phenotype was sex-dependent, we performed a 2-way ANOVA (genotype $\times$ sex). The analysis revealed a significant main effect of genotype ($P < 0.05$) with no significant interaction ($P > 0.05$), confirming that the $\approx 70\%$ reduction was consistent across both sexes (Figure S14). This reduction remained statistically significant and without sex interactions, whether data were analyzed as absolute counts or normalized as a ratio to CD45. As a result, we combined male and female data for a more robust analysis, and confirmed a $\approx 70\%$ reduction in CD8 T cells, despite similar levels of CD45 and CD11b cells $\approx 70\%$ (Figure 4D and 4E; $P < 0.01$). Immunostaining of plaque revealed CD8 T-cell infiltration into intimal lesions of the innominate artery at 9 months, not observed to the same extent in EC knockout lesions (Figure 4F; Figure S15). To examine T-cell phenotypes in the evolved plaque, we performed a cellular indexing of transcriptomes and epitopes by sequencing analysis of plaque from 9-month-old chronic hypercholesterolemic mice. In the blood, we confirmed similar levels of T cells (Figure S16), and a trend toward reduced levels in plaque tissues (67% reduction in mean, $P = 0.1$; Figures S17 and S18). Plaque T cells in ECKO mice showed reduced signs of activation (Figure S18). Although we cannot exclude similar effects on CD4 T cells, their limited quantities in this model do not

Figure 2 Continued. C1qa in the low and disturbed flow (LDF) and contralateral arteries of WT and ECKO mice. Each point is an individual artery from a unique mouse. **G**, Gene set enrichment analysis of pathways affected in total mRNA and TRAP mRNA between WT and ECKO mice, showing significantly enriched Gene Ontology (GO) terms, clustered into major categories. Inset shows an example of the underlying data for each square on the heatmap. **B**, *** $P < 0.0001$, ANOVA with post hoc Tukey test. **F**, *** $P < 0.0001$, DEseq2. **G**, *False discovery rate (FDR) < 0.05 , **FDR < 0.01 , ***FDR < 0.001 by gene set enrichment analysis-weighted analysis. **B**, Scale bar = 20 μ m. IL indicates interleukin; TCR, T-cell receptor; and Tpm, transcripts per million.

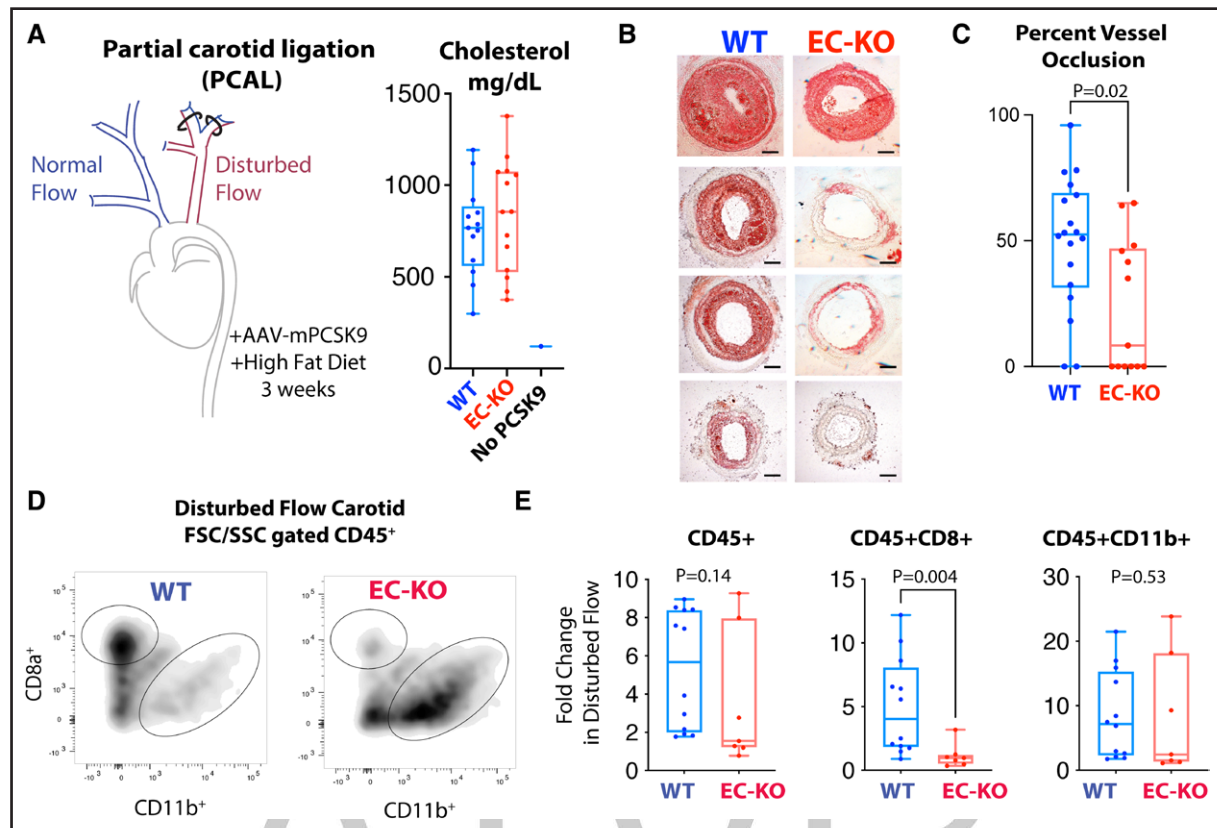


Figure 3. Endothelial deletion of Elavl1 reduces CD8 T-cell accumulation in the acute plaque model.

A through **C**, Analysis of atherosclerotic plaque in ligated vessels 3 weeks post-partial carotid artery ligation (PCAL); $n=18$ wild-type (WT; 7 male, 11 female), $n=12$ endothelial cell (EC)-knockout (KO; 4 male, 8 female). **A**, Schematic of PCAL model. **B**, Examples of Oil Red O-stained cross-sections of the partially ligated left carotid artery. Sections were taken from the mid-section of each carotid, to avoid complications of flow near branch points. **C**, Quantification of plaque as percent vessel occlusion, from sections taken at the midpoint of each excised disturbed flow carotid. Each point is an individual artery from unique mouse. Contralateral normal flow arteries showed no plaque. **D**, Flow cytometry plots of size-gated CD45 cells isolated from the disturbed flow carotid. **E**, Fold increase in the disturbed flow artery vs contralateral normal flow control in the number of CD45+, CD45+CD8+, and CD45+CD11b+ cells. $n=12$ WT (4 male, 8 female), $n=7$ EC-KO (4 male, 3 female). Each point is the result of the flow cytometry comparison of disturbed flow and contralateral normal flow arteries from a unique mouse. **C** and **E**, * $P<0.05$, ** $P<0.01$ by Mann-Whitney U test. **B**, Scale bar=100 μ m. PCSK9 indicates proprotein convertase subtilisin/kexin type 9.

allow us to make definitive conclusions. Thus, accumulation of activated CD8 T cells is diminished in plaque tissues of ECKO mice, despite little discernible effect on the circulating CD8 T-cell population.

Endothelial Elavl1 Is a Stromal Checkpoint for CD8 T-Cell Persistence in Plaque

Plaque phenotypes suggested that the deletion of Elavl1 from the endothelium inhibits T-cell accumulation, and transcriptional data suggest that this may be due to effects on antigen response and activation, rather than recruitment. Activation of CD8 T cells in vivo and analysis of their accumulation in atherosclerotic plaque in the PCAL model following our published protocol¹⁸ confirmed that there was no deficit in activated T-cell recruitment to lesions (Figure S19). Therefore, to assess effects on T-cell persistence after antigen stimulation, we developed an in vitro system,

coculturing the cell types necessary to recapitulate the phenotype seen in vivo: ECs, antigen-presenting bone marrow-derived dendritic cells (BMDCs), and CD8 T cells. We generated ROSA-CreERT2 ECs with floxed Elavl1 to allow us to excise Elavl1 in vitro by treatment with tamoxifen metabolite 4-hydroxytamoxifen. As atherosclerotic plaque antigens are not yet well defined, we used genetically engineered OT-I CD8 T cells specific to processed OVA-derived peptide presented via MHC-I on BMDCs. Binding of the OT-I CD8 T-cell receptor to MHC-I-OVA peptide on the BMDC (signal 1) activates T-cell expansion and survival, provided sufficient costimulatory (signal 2) and survival cues (signal 3) are present. CD8 T-cell were marked by CD45.1 and APCs with CD45.2, allowing us to separate these cells from each other and the cocultured CD45-negative ECs by flow cytometry (Figure 5A and 5B). In this system, we tested the effect of ECs with and without deletion of Elavl1 on T-cell persistence.

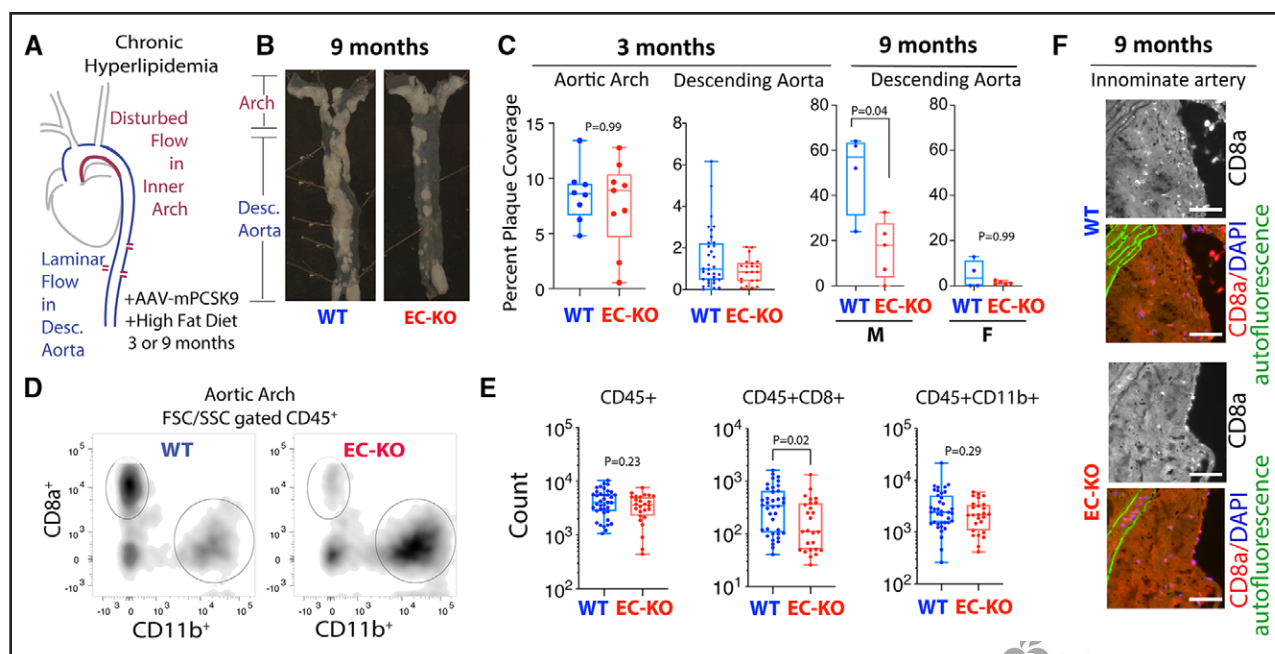


Figure 4. Endothelial deletion of *Elavl1* reduces CD8 T-cell accumulation in the aortic arch of hypercholesterolemic mice.

A through **E**, Analysis of atherosclerotic plaque in aortas of mice collected after 3 months or 9 months of hyperlipidemia. **B**, Representative images of male aortas at 9 months of hyperlipidemia. **C**, Quantification of plaque as percent vessel coverage in the aortic arch and descending aorta. Males and females are separated in the 9-month data; only minimal plaque was observed in females at this time point. Each point is an individual artery. At 3 months, $n=31$ wild-type (WT; 19 male, 12 female), $n=19$ endothelial cell (EC)-knockout (KO; 15 male, 14 female), and at 9 months, $n=8$ WT (4 male, 4 female), $n=10$ EC-KO (5 male, 5 female). **D** and **E**, Flow cytometry plots and quantitation of the aortic arch after 3 months of hyperlipidemia. $n=38$ WT (19 male, 19 female), $n=26$ EC-KO (11 male, 15 female). Each point is the result of the flow cytometry analysis from a single mouse. **F**, Immunofluorescence staining of CD8 cells in the innominate artery of 9-month old EC-KO mice and WT control. Single channel shows CD8 staining, and merged channel includes DAPI (4',6-diamidino-2-phenylindole) and Alexa488 channel autofluorescence to show elastin. **C** and **E**, * $P<0.05$, ** $P<0.01$ by Mann-Whitney U test. **F**, Scale bar=50 μ m.

T-cell response to antigen typically peaks between 7 and 15 days,⁴³ so T-cell proliferation, and survival were examined after 13 days in coculture. As expected, there was a marked increase in CD8 T-cell number (≈ 2.5 -fold, $P<0.001$) in response to full-length antigen Ova when cocultured with same-sex WT ECs and BMDCs (Figure 5B and 5C). However, deletion of *Elavl1* in ECs entirely blocked this response, despite

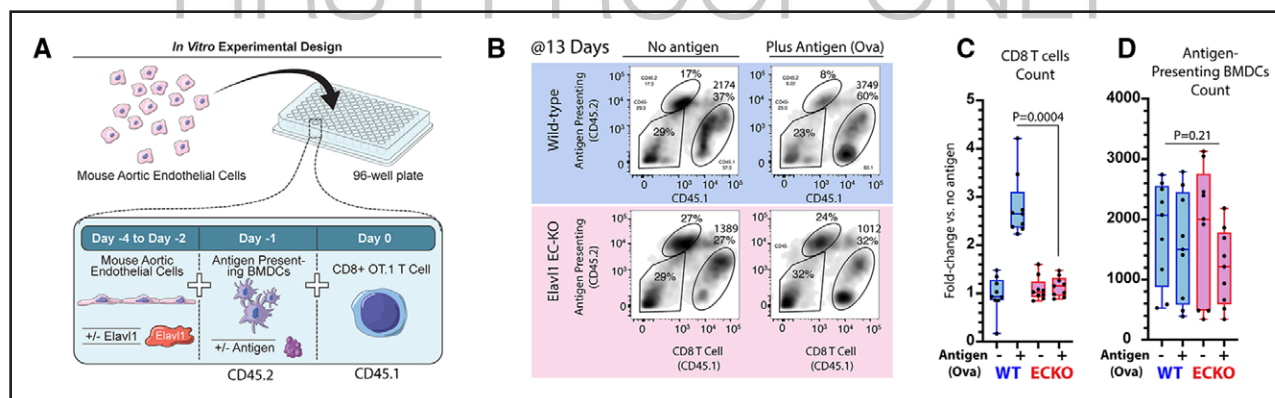


Figure 5. Deletion of *Elavl1* from endothelial cells (ECs) reduces long-term persistence of CD8 antigen responses in vitro.

A, Schematic of EC and bone marrow-derived dendritic cells (BMDCs) and CD45.1 major histocompatibility complex type I class-restricted, OVA (ovalbumin)-specific transgenic mice (OT-I) CD8 T-cell coculture in vitro. **B**, Example flow cytometry plots of CD45.1 OT-I T cells, after 13 days of coculture. **C** and **D**, Quantitation of CD45.1 CD8 OT-I T cells and CD45.2 BMDCs after 13 days of coculture with or without cognate antigens (**C**) OVA (ovalbumin; 0.5 μ g/mL OVA, in media throughout coculture) or (**D**) SIINFEKL (processed peptide, antigen for OT-I cells; 1 μ g/mL, preincubation with ECs and BMDCs). **C** and **D**, Each point represents results from an individual well. **C** and **D**, $n=9$ vs 9, 2 separate experimental dates, 5 experimental dates. **C** and **D**, Data normalized to no antigen, wild-type (WT) ECs (except raw count data for BMDCs, which is not normalized). ECKO indicates postnatal endothelial-specific *Elavl1*/HuR deletion; and KO, knockout. Comparisons by ANOVA with Sidak multiple comparisons test, **** P adjusted <0.0005 .

the same number of WT BMDCs in the system available to present antigen (Figure 5B and 5C). This was not a result of reduced numbers of APCs (Figure 5D) or a generally impaired response to cytokine, because pulsing of BMDCs with the processed SIINFEKL peptide of OVA before CD8 T-cell addition elicited similar responses on both EC types (Figure S20). To examine the kinetics, we collected cocultured cells after 3 days, and found a significant increase in CD8 T-cell counts on WT ECs and a lesser and nonsignificant increase on ECKO ECs (Figure 6A and 6B). However, the largest observable defect was in the viability of T cells on ECKO cells, which sharply declined (Figure 6C). Again, this was not due to a significant difference in the BMDCs (Figure 6D; Figure S21). Interestingly, protein levels of C1q were increased specifically in ECKO ECs in this coculture, in both supernatant and the extracted protein of ECs after blocking protein secretion (Figure S22). Prior work has shown that C1q expression in stromal cells, or exogenously added to CD8 T cells, limits expression to mild antigen stimulation, but not SIINFEKL.⁴⁴ Thus, the data show that despite an equivalent ability to respond to an unprocessed antigen (SIINFEKL), CD8 T cells cultured on ECKO ECs are impaired in survival and persistence in response to processed antigen (OVA).

These findings suggest the increased Elavl1 activity in the atherosclerotic endothelium acts as a checkpoint to enable CD8 T-cell persistence.

DISCUSSION

Here, we demonstrate that Elavl1, which is upregulated in ECs in the atherogenic setting of disturbed flow, suppresses a switch in the ribosomal association of mRNA and T-cell persistence in the artery wall. Mechanistically, this includes the altered ribosomal association or splicing of transcripts with the potential to regulate T-cell response and leads to reduced T-cell survival on Elavl1-deleted endothelium cells in vitro, despite the presence of APCs and cognate antigen. In vivo, endothelial-specific deletion of Elavl1 reduces CD8 T-cell numbers and markers of activation in atherosclerotic lesions without substantial effects on circulating T-cell numbers or the recruitment of activated T cells to lesions. Thus, Elavl1 low ECs are capable of actively and locally repressing immune responses, mimicking properties of regulatory T cells and tolerizing dendritic cells.⁴⁵

Endothelial Elavl1 and the Persistence of CD8 Effector Memory Cells in Plaque

Our data show that endothelial Elavl1 is required for the persistence of activated CD8 T cells in atherosclerotic lesions. The specific antigens in atherosclerotic lesions are likely diverse; there is evidence that T cells in atherosclerotic plaque may have been activated by viral infections, oxidized apolipoproteins, and autoimmune

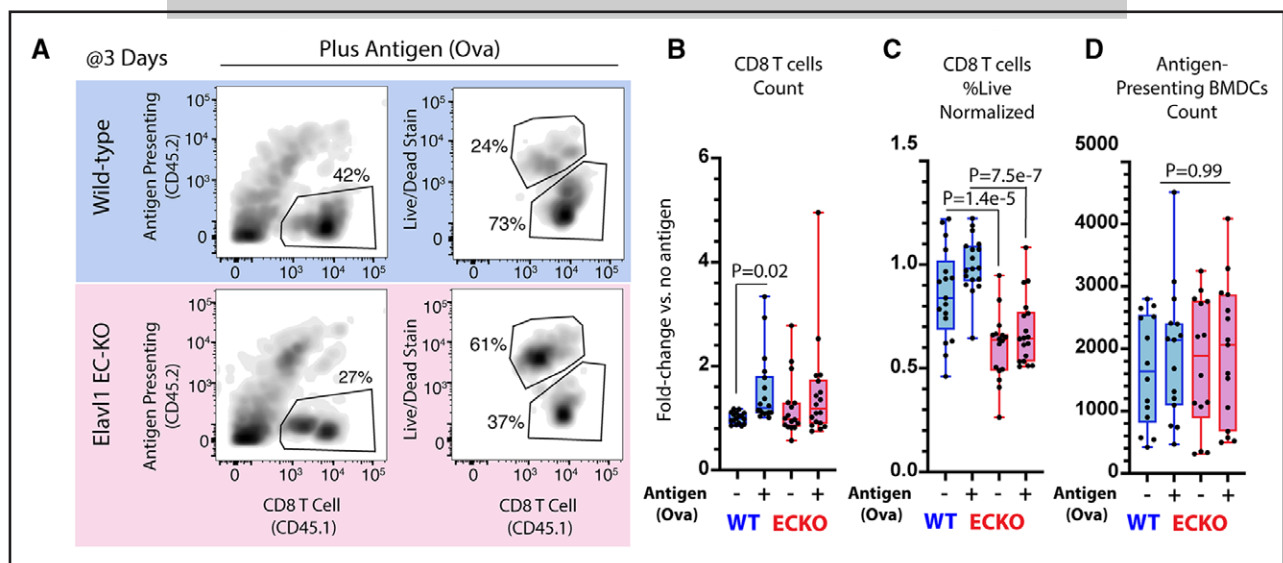


Figure 6. Endothelial deletion of *Elavl1* reduces survival of cocultured CD8 T cells in vitro.

Results, by quantitative flow cytometric analysis, of changes in cell abundance in an in vitro culture of endothelial cells (ECs), antigen-presenting cells (CD45.2) and CD8 major histocompatibility complex type I class-restricted, OVA (ovalbumin)-specific transgenic mice (OT-I) T cells (CD45.1), with or without cognate antigens OVA (0.5 µg/mL OVA, in media throughout coculture). **A**, Example flow cytometry plots after 3 days of coculture, showing live/dead gating of CD8 T cells. **B**, Quantitation of CD45.1 CD8 OT-I T cells, **(C)** percent live CD8 OT-I T cells, and **(D)** CD45.2 bone marrow-derived dendritic cells (BMDCs). **B** through **D**, Each point represents results from an individual well, N>16 each, 5 experimental dates. **B** through **D**, Data normalized to no antigen, wild-type (WT) ECs (except raw count data for BMDCs, which is not normalized). ECKO indicates postnatal endothelial-specific Elavl1/HuR deletion; and KO, knockout. Comparisons by ANOVA with Sidak multiple comparisons test, ***P* adjusted [Padj] <0.005, *****P*adj<0.0005.

disease.¹⁷ Regardless of their initial stimulation, once recruited to plaque, CD8 T cells persist for months at least as effector memory cells (CD8 T effector memory cells).^{15,18} Consistent with enrichment of CD8 T cells with antigen-specific response, human coronary arteries have shown high clonality in plaque CD8 T cells, which increased according to the stage of the plaque.⁴⁶ Our findings show how the stroma, and the endothelium in particular, contributes to the local maintenance of CD8 T effector memory cells. Because Elavl1 expression is potently induced by platelet interactions with the arterial wall, it suggests that Elavl1 induction in the endothelium may be an important checkpoint towards T-cell persistence in plaque. As a stromal checkpoint, this would allow for local modulation of immune responses, as we show here, through effects on plaque recruited but not circulating levels of CD8 T cells. Approaches targeting local checkpoints are likely to provide important advantages for local immune-suppression without many of the consequences of systemic immunosuppression.

Translational Regulation in Atherosclerosis

Here, we show that Elavl1 limits immunosuppressive changes in mRNA translation in the atherogenic endothelium. Translation of mRNA is a tightly regulated process, and variations between mRNA isoforms can lead to >100-fold differences in protein translation.⁴⁷ However, very little is known about the regulation of translation during chronic endothelial activation in atherosclerotic lesions. In myeloid cells and the endothelium, Elavl1 has principally been considered in the context of 3'UTR binding and transcript stability.^{48,49} Given this emphasis in the literature, it was somewhat surprising to us that the major effect we observed was not on canonical Elavl1 stabilized transcripts. Such analysis may require additional *in vivo* experiments to examine transcript kinetics. Nevertheless, whatever these effects may be, they do not seem to strongly affect canonically stabilized transcripts (eg, *Ifit5*, *Nfkb1*, *Nfkb2*, *Ifitm3*, *Irf9*, *Mapk1*, *Ifnb1*, *Ptgs2/Cox-2*, *Tgfb*, *Ctss*, *Sele*)^{48,50} in either the total mRNA or TRAP mRNA pool. The finding that Elavl1 can instead regulate ribosomal binding and protein translation is new in this context but has been reported in epithelial cells and B cells. In hypoxic epithelial cells, an unbiased proteomic analysis revealed a 3- to 4-fold enrichment of Elavl1 at the polysome.²⁴ Previous analysis of the effect of Elavl1 loss in the endothelium had revealed alterations in the splicing of the transporter of a key translational protein, eIF4e transporter (*Eif4enif1*), which we confirmed.⁵¹ In future work, it will be interesting to examine whether Elavl1 effects on T-cell responses are coupled with cell metabolic alterations. Supporting this idea, *Hif2a* enhanced polysome association of Elavl1 in epithelial cells,²⁴ and *Hif2a* levels are increased in ECs exposed to disturbed flow in the absence of hypoxia.⁵² Thus, changes

in ribosomal association of specific mRNA and the RBPs that regulate this process may be essential in the coupling of metabolic dysfunction and the immune response.

Together, our work shows that changes in the ribosome-bound translome in atheroprone endothelium are concentrated in genes involved in the regulation of chronic immune-endothelial interactions, defining new targets to modulate chronic immune-endothelial interactions. The RBP Elavl1 affects this response and functions as a novel stromal checkpoint to T-cell persistence in atherosclerotic lesions. We predict that this may be important not only in the accumulation of CD8 effector cells in plaque, as we show here, but also likely in other settings of chronic immune responses, including autoimmune disease and graft-vs-host responses, and in settings where effector cell accumulation is impaired, as in cancer immunotherapy and antiviral responses in old age. The endothelium, found in all vertebrates, evolutionarily predates the appearance of T cells. It is perhaps not surprising, then that as the adaptive immune system developed, it was entwined with endothelium. Therefore, we expect that this is just the beginning of work to understand how these interactions locally modify immune cell function.

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Disclosures

A. Rodriguez-Oquendo is the founder of Lipid Genomics. The other authors report no conflicts.

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